

Assignment 1

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June 17, 2026

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1 Introduction to the Newsvendor model

Suppose we want to sell a product, and we need to choose the quantity to produce (or to order from a supplier), assuming the production window precedes the sales window. Suppose also that we adopt the following notation:

- q : quantity produced/ordered;
- c : unit production/ordering cost;
- D : demand observed during the sales window;
- p : unit price during the sales window;
- r : discounted price, for any unsold products during the sales window.

The best choice would be to opt for a quantity q equal to the demand D ; however, the latter is not known a priori, as the product order precedes the actual realization of the demand.

As a direct consequence of this, disregarding the ideal scenario where $q = D$, two situations could occur:

- $q < D$: producing/ordering a quantity lower than the demand that actually occurs implies that a portion $D - q$ of this demand remains unsatisfied (*stockout*). This leads to an opportunity cost (also known as *unit cost of underage*)

$$c_u = p - c \quad (1)$$

Similarly, it can be understood as the lost profit for each unit of unsatisfied demand.

- $q > D$: in the opposite case, i.e., overproduction compared to the actual demand, the surplus of products will have to be sold off. This leads to a cost (*unit cost of overage*)

$$c_o = c - r \quad (2)$$

The situation just described represents the newsvendor problem.

1.1 Expected profit maximization

The approach we apply below to determine the quantity q involves modeling the demand as a continuous random variable with probability density function $f_D(x)$ and maximizing the expected value of the profit.

Starting from the profit equation

$$\pi(q, d) = c_u \cdot \min(q, d) - c_o \cdot (q - d)^+ \quad (3)$$

we set up the calculation for its expected value

$$\mathbb{E}[\pi(q, D)] = c_u \left(\int_0^q x f_D(x) dx + \int_q^{+\infty} q f_D(x) dx \right) - c_o \int_0^q (q - x) f_D(x) dx \quad (4)$$

where x is a dummy variable of integration representing the demand. Then, using Leibniz's rule, we derive the derivative of the function

$$\frac{d\mathbb{E}[\pi(q, D)]}{dq} = c_u \cdot (1 - F_D(q)) - c_o \cdot F_D(q)$$

where $F_D(x) = P\{D \leq x\}$ represents the cumulative distribution function of the demand. By equating the derivative to zero¹, we obtain the following result

$$F_D(q^*) = \frac{c_u}{c_u + c_o} \quad (5)$$

In other words, the optimal quantity q^* to order to maximize the expected profit is a quantile of the demand distribution. For this reason, the ratio $\frac{c_u}{c_u + c_o}$ is sometimes also known as the *level of service*, and it represents the probability of not experiencing a stock-out.

1.2 Normal demand hypothesis

For the case study we analyze below, we assume a Normal distribution for the demand

$$D \sim N(\mu, \sigma^2)$$

This hypothesis allows us to derive the equation for the optimal quantity

$$q^* = \mu + z \left(\frac{c_u}{c_u + c_o} \right) \cdot \sigma \quad (6)$$

and the expected profit as

$$\mathbb{E}[\pi(q, D)] = c_u \cdot \mu - c_o(q - \mu) - (c_u + c_o) \cdot \sigma \cdot \mathcal{L} \left(\frac{q - \mu}{\sigma} \right) \quad (7)$$

with $\mathcal{L}$, the *loss function*, expressed in the following form (ϕ = pdf, Φ = cdf)

$$\mathcal{L}(z) = \phi(z) - z(1 - \Phi(z)) \quad (8)$$

In the following sections, to prevent the demand from taking negative values, we will use a standard deviation value that is sufficiently small compared to the expected value. In this way, the probability of observing negative demand can be considered negligible.

¹the second derivative is negative (concavity), therefore the result actually refers to a maximum

2 Simulation Parameters

Through Monte Carlo simulations, we aim to analyze the impact of the uncertainty of some parameters involved on the estimation of the optimal quantity and the consequent expected profit. More precisely, the reference metric is the ratio between the expected value of the profit actually obtained by ordering/producing sub-optimal quantities $\hat{q}$ of the product, and what would be obtained in the absence of parameter uncertainty, i.e., by ordering q^* .

$$EP_{\text{ratio}} = \frac{\mathbb{E}[\pi(\hat{q}, D)]}{\mathbb{E}[\pi(q^*, D)]} \quad (9)$$

The ratio approaches 1 the closer the sub-optimal quantity $\hat{q}$ is to q^* .

Unless otherwise specified in particular cases (which we will analyze later), we conduct the analyses under the following assumptions:

- economic parameters: $p = 100$, $c = 60$, $r = 45$, $c_u = 40$, $c_o = 15$ (all expressed in \$).
- true parameters of the demand distribution: $\mu_{\text{true}} = 1000$, $\sigma_{\text{true}} = 150$
- Monte Carlo runs: $M = 10,000$ replications

Furthermore, for simplicity, we assume that $q \in \mathbb{R}_+$, meaning we hypothesize it is possible to order/produce fractions of the product (ideally q could be in tons, cubic meters ...) using at most one decimal place.

Following the choices above, we can immediately derive the optimal theoretical quantity to produce/order using equation (6), and the consequent optimal expected profit at q^*

$$q^* = 1090.7 \text{ units} \quad , \quad \mathbb{E}[\pi(q^*, D)] = 37258.5\$$$

3 Impact of sample size

A primary source of uncertainty, which produces a sub-optimal solution, lies in the estimation of the demand distribution parameters. In this section, we analyze how the estimates of the sample mean and standard deviation, $\hat{\mu}$ and $\hat{\sigma}$, change as the size N of the used demand sample varies. The goal is to observe how the use of $\hat{\mu}$ and $\hat{\sigma}$ instead of population parameters affects the estimate of the quantity to be produced/ordered, and consequently the expected profit ratio.

More in detail, given a fixed N :

1. We generate $M = 10,000$ possible demand scenarios, outputting a matrix of size $M \times N$. Each row of the matrix represents a possible demand scenario, and the length of this row is exactly equal to the sample size.
2. For each of the M generated scenarios, we calculate the sample mean and standard deviation

$$(\hat{\mu}_i, \hat{\sigma}_i), \quad \forall i = 1, \dots, M$$

3. For each pair of parameters, we calculate the corresponding (sub-)optimal quantity $\hat{q}$ using equation (6), to obtain a distribution of possible values $(\hat{q}_1, \dots, \hat{q}_M)$.
4. Starting from the quantity vector determined in the previous step, using equation (9) we calculate the ratio vector, keeping the theoretically optimal quantity fixed in the denominator and computing the i -th expected profit $\mathbb{E}[\pi(\hat{q}_i, D)]$ from equation (7).

To observe the effect of N , we tested $N = [2, 5, 10, 20, 30, 50, 200, 500, 1000]$. The choice to opt for relatively small samples is not random; on the contrary, it is intended to simulate real-world situations of data scarcity as closely as possible.

3.1 Simulation results

The simulation results are graphically summarized in two violin² plots (Fig. 1) and quantified in a table (Tab 1).

- In the left graph, which illustrates the distribution of the estimated sub-optimal quantity $\hat{q}$ as N varies, we observe a very pronounced parametric uncertainty for demand scenarios with few observations. In fact, in particularly unlucky cases, we would end up ordering quantities $\hat{q}$ that are even several hundred units away from the optimum q^* . As the sample size increases, the uncertainty progressively decreases, and the distribution collapses towards q^* .
- In the right graph, we observe the corresponding distributions of the EP_{ratio} ratios (9). They show a marked asymmetry, particularly evident for small values of N , as a direct consequence of the geometry of the expected profit function $\mathbb{E}[\pi(q, D)]$: this function is concave and reaches its absolute maximum at q^* . Consequently, the distribution of the EP_{ratio} is bounded above by 1 by construction (no sub-optimal estimate can generate a profit higher than the theoretical one), while its left tail can extend freely downwards, generating the asymmetry visible in the graphs.

This phenomenon is amplified for small N : in particularly unlucky cases, the expected profit obtained would be reduced by more than 30% compared to the theoretical maximum.

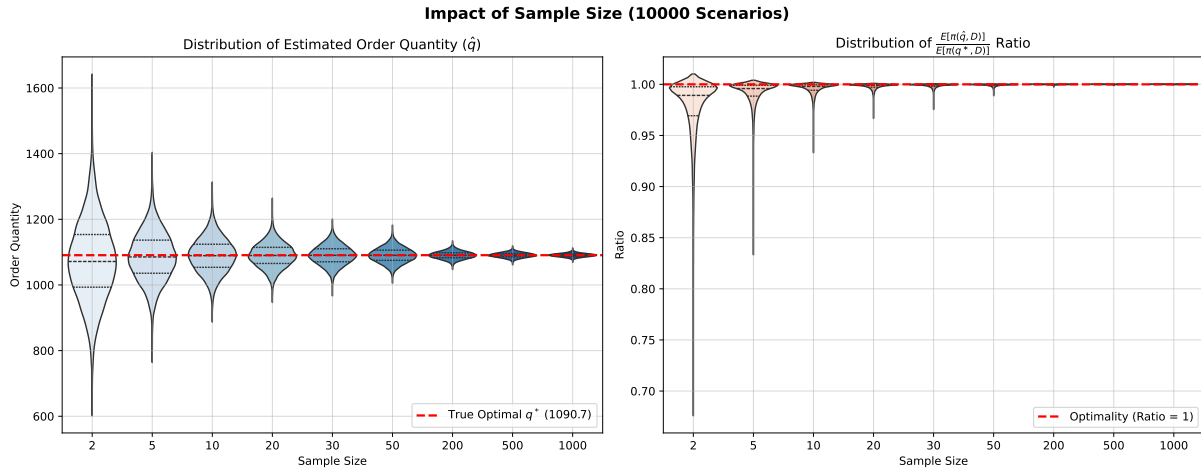


Figure 1: Impact of sample size on ordered quantity (left) and expected value ratio (right)

N	Mean	Std	Median	2.5%	97.5%	95% CI Width	% Ratio > 1
2	0.9770	0.0319	0.9892	0.8869	1.0000	0.1130	0.0000
5	0.9913	0.0126	0.9959	0.9561	1.0000	0.0439	0.0000
10	0.9957	0.0060	0.9980	0.9784	1.0000	0.0215	0.0000
20	0.9978	0.0031	0.9990	0.9890	1.0000	0.0110	0.0000
30	0.9986	0.0020	0.9994	0.9931	1.0000	0.0069	0.0000
50	0.9991	0.0012	0.9996	0.9957	1.0000	0.0043	0.0000
200	0.9998	0.0003	0.9999	0.9989	1.0000	0.0011	0.0000
500	0.9999	0.0001	1.0000	0.9996	1.0000	0.0004	0.0000
1000	1.0000	0.0001	1.0000	0.9998	1.0000	0.0002	0.0000

Table 1: Impact of the sample size (N) on the expected profit ratio distribution.

Although the tails of the distribution for very small N potentially risk eroding the profit (at least in expected value), this effect becomes gradually less pronounced by increasing the number of observations to modest values. As evidence of this, from the summary table we can observe that for $N = 10$ the width of the confidence interval is reduced by a full 5 times compared to the $N = 2$ case.

²preferred over the box plot, as it provides a better perception of the distribution tails

4 Impact of estimation error on the salvage price

Often, the discounted price r is not known with certainty at the time of the order, meaning we do not know exactly at what price we will be able to sell any unsold products. The reasons for this uncertainty are varied, and depend on the context (market fluctuations, product degradation...)

In this section, we analyze how an estimation error on this parameter propagates to the ordered quantity and the expected profit. We model the estimated salvage value $\hat{r}$ as a random variable distributed according to a Truncated Normal:

$$\hat{r} \sim \text{TruncNorm}(r, \sigma_r, a, b) \quad (10)$$

centered on the true value $r = 45\$$, with variable standard deviation σ_r . The motivations behind this choice are essentially the following:

- the salvage price cannot be negative;
- the salvage price must be strictly less than the unit cost $c = 60\$$. If this were not the case ($r \geq c$), an arbitrage opportunity would arise, since it would be profitable to order an infinite quantity of the product, guaranteeing a risk-free profit.

Therefore, we set the distribution parameters to $(a, b) = (0, c - \varepsilon)$.

To evaluate the impact of an uncertain discounted price, we perform a simulation with $M = 10,000$ scenarios, testing different levels of uncertainty $\sigma_r \in [1, 2, 5, 10, 15]$. For each scenario, the estimate $\hat{r}$ distorts the overage cost ($\hat{c}_o = c - \hat{r}$), altering the expected level of service (5) and producing a sub-optimal estimated quantity $\hat{q}$.

4.1 Simulation results

The results are shown in the violin plots in Figure 2 and quantified in Table 2.

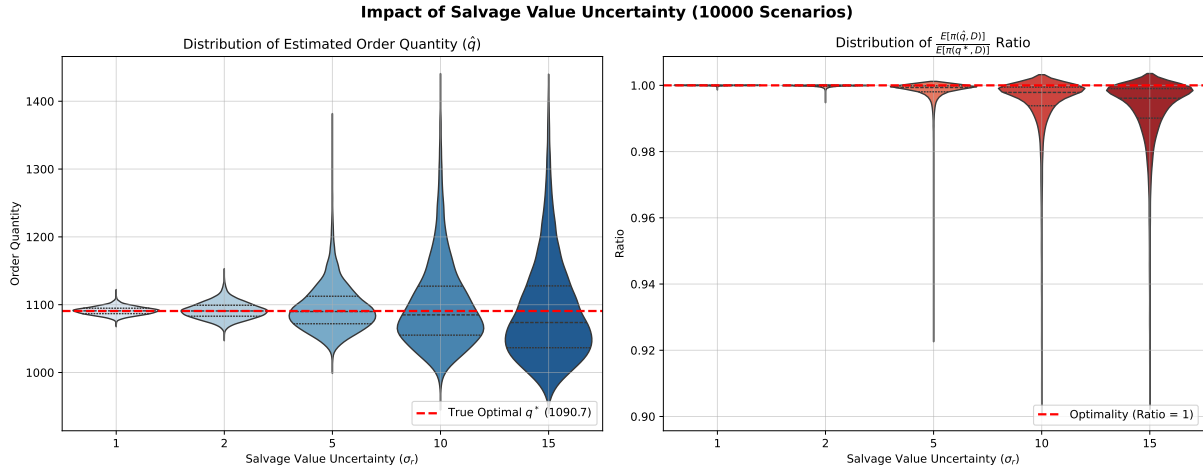


Figure 2: Impact of salvage price uncertainty on ordered quantity (left) and profit ratio (right).

σ_r	Mean	Std	Median	2.5%	97.5%	95% CI Width	% Ratio > 1
1	0.9999	0.0001	1.0000	0.9997	1.0000	0.0003	0.0000
2	0.9998	0.0004	0.9999	0.9988	1.0000	0.0012	0.0000
5	0.9982	0.0038	0.9993	0.9892	1.0000	0.0108	0.0000
10	0.9944	0.0102	0.9979	0.9629	1.0000	0.0371	0.0000
15	0.9922	0.0113	0.9961	0.9589	1.0000	0.0410	0.0000

Table 2: Impact of salvage price uncertainty (σ_r) on the expected profit ratio distribution.

Analyzing the graphs, two key considerations emerge:

- In the left graph, we immediately observe how, as the uncertainty σ_r grows, the tails of the estimated sub-optimal quantities distribution become increasingly pronounced. Furthermore, it is interesting to observe how the distribution of $\hat{q}$ presents an asymmetry with more pronounced tails towards values $\hat{q} > q^*$; the phenomenon becomes more marked for high σ_r . This happens because for large values of $\hat{r}$, close to the upper limit $c - \varepsilon$ (which we visit more often in high-uncertainty scenarios σ_r), the overage cost (2) tends to zero. As a direct consequence, the level of service tends to 1, which translates into a solution $\hat{q}$ such that the probability of stock-out is close to zero: this results in very high quantities $\hat{q}$.
- In the right graph, we observe the repercussions of an uncertain discounted price on the EP_{ratio} (9). Looking at the y-axis and referring back to the table: even for situations of extreme uncertainty (e.g. $\sigma_r = 15$), particularly unlucky scenarios produce a deterioration of the expected profit compared to the optimal theoretical one of about 4% (2.5% quantile ≈ 0.96), which is far lower than the repercussions found in parametric uncertainty in the previous section.

5 Impact of the c_u/c_o ratio

In this section, we shift the analysis towards the economic side more markedly. From section 1, we know the meaning of c_u and c_o . Consequently, we can state that:

- if $c_u/c_o > 1$, it follows that the gain from selling one unit of product exceeds (the more the ratio is greater than 1) the cost we must bear when a product remains unsold during the sales window (and we find ourselves having to dispose of it at a price r). When this happens, the level of service (5) grows and the stock-out probability decreases. Ultimately, the model prefers the purchase of a quantity q^* higher than the average demand μ .
- if $c_u/c_o < 1$, exactly the opposite happens. Since the cost to bear in the presence of unsold products is greater than the unit profit, the model prefers the purchase of a quantity q^* lower than the average demand μ .

To have a better overview, we rewrite the level of service by making the ratio $\rho = \frac{c_u}{c_o}$ explicit:

$$los = \frac{c_u}{c_u + c_o} = \frac{\rho}{\rho + 1} \quad (11)$$

Starting from these considerations, the objective of this section is the following: we want to understand in which economic setting parametric uncertainty (using $(\hat{\mu}, \hat{\sigma})$ instead of (μ, σ)) is more harmful. In accordance with the previous sections, a scenario is more "harmful" to us the further the EP_{ratio} strays from the unitary theoretical value.

5.1 Simulation results

Regarding the parameters used, we refer to those initially set in section 2, except for p , the unit price during the sales window. In other words, we choose to vary the ratio ρ by acting solely on p and leaving everything else unchanged. This choice is the closest to a real scenario, where a company positions itself in an economic setting based precisely on the retail price p of the product.

The results of the analysis are visually illustrated in the graphs in Figure 3. In Table 3, for the sake of brevity, we report only the detailed statistics for the $N = 10$ case, but the attached `Python` code performs the calculations for the other sizes as well. Regarding the heat map, we use the 5th percentile of the EP_{ratio} ratio as a tail risk measure: it represents the guaranteed profit threshold in 95% of the simulated scenarios.

We can make the following observations:

- From the left violin plot, assuming demand scenarios with a reduced number of data points ($N = 10$), we observe that the left tails of the distribution (i.e. moving away from the optimum) are more marked the lower the ratio ρ is.

In other words, parametric uncertainty is significantly more impactful for economic settings where a surplus is inconvenient ($c_u < c_o$). Looking at the table, the EP_{ratio} even drops below 80% (2.5% quantile equal to 0.7918).

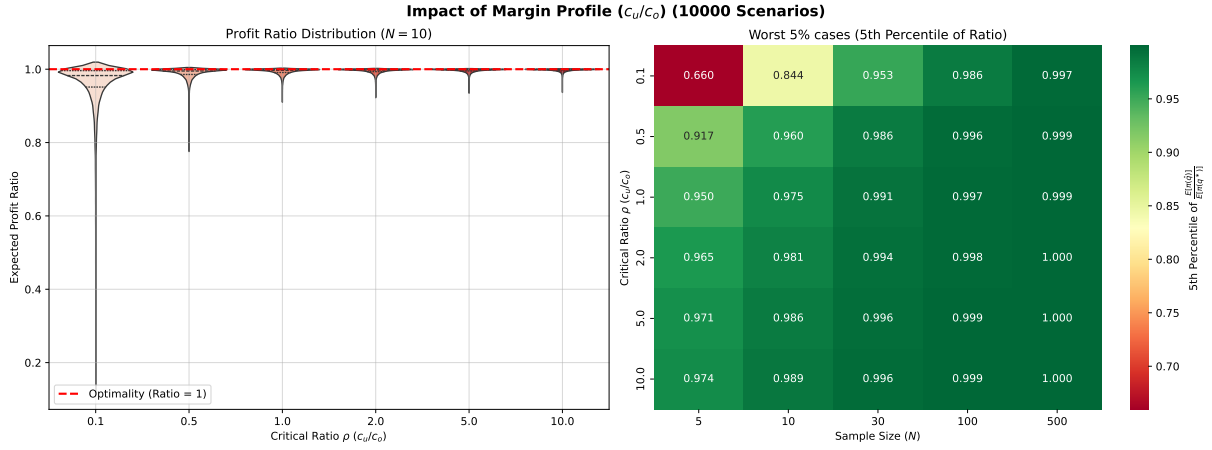


Figure 3: Profit ratio distribution for a fixed sample ($N = 10$) as ρ varies (left). Heat map of the 5th percentile as N and ρ vary (right).

ρ	Mean	Std	Median	2.5%	97.5%	95% CI Width
0.1	0.9607	0.0609	0.9827	0.7918	1.0000	0.2082
0.5	0.9894	0.0148	0.9951	0.9492	1.0000	0.0508
1.0	0.9934	0.0091	0.9970	0.9669	1.0000	0.0331
2.0	0.9950	0.0070	0.9977	0.9754	1.0000	0.0246
5.0	0.9963	0.0055	0.9983	0.9810	1.0000	0.0190
10.0	0.9971	0.0045	0.9988	0.9843	1.0000	0.0157

Table 3: Summary statistics of the impact of the margin profile ($\rho = c_u/c_o$) on the profit ratio, setting the sample size $N = 10$.

- From the right heat map we have a complete picture of the phenomenon. The pattern highlighted in the previous point for fixed N remains: the smaller ρ is, the more parametric uncertainty causes the EP_{ratio} to stray from the theoretical optimum value of 1. In the worst case of small N and ρ (e.g. $N = 5$, $\rho = 0.1$), in the worst 5% of cases we realize barely 66% of the optimal profit. However, from the heat map we can make an additional observation: as the demand sample size increases, the estimation of $\hat{\mu}$ and $\hat{\sigma}$ improves significantly. As a direct consequence of this, parametric uncertainty is reduced, and even for small values of ρ the EP_{ratio} approaches the theoretical optimum.

From an economic point of view, we can draw some conclusions from the results just commented. Considering all the simplistic assumptions made, for products with a low profit margin and high cost for unsold items, it is crucial to perform an accurate estimation of the distribution parameters, in order to order/produce a quantity close to q^* and thus obtain an expected profit as close as possible to the theoretical maximum. As already seen in section 3, for this purpose it is necessary to intervene on the sample size N .

Conversely, for products with a high profit margin, the accuracy of parametric estimation has a limited impact, thus allowing for demand scenarios with a reduced number of data points.

6 Impact of the coefficient of variation

The final source of uncertainty we analyze concerns the intrinsic volatility of demand, measured by the coefficient of variation

$$CV = \frac{\sigma}{\mu} \quad (12)$$

More precisely, we want to understand what the effect of an increasingly unpredictable demand (increasing σ) is on the EP_{ratio} . To do this, during the simulation we keep all parameters fixed and only vary σ , the standard deviation of the demand population. For completeness of analysis, we repeat the simulation for different values of N .

6.1 Theoretical considerations

Before presenting the simulation results, it is useful to reason about the expected behavior of the EP_{ratio} ratio as CV varies. Substituting the optimal quantity (6) into the expected profit equation (7), we obtain

$$\mathbb{E}[\pi(q^*, D)] = c_u \mu - \sigma [c_o \cdot z(\text{los}) + (c_u + c_o) \cdot \mathcal{L}(z(\text{los}))] \quad (13)$$

All else being equal, under the Normal demand hypothesis, an increase in σ reduces the optimal expected profit.

Regarding the numerator of the EP_{ratio} , the reasoning is practically analogous: as σ varies, the ratio would be consistently unitary if it were not that $\mathbb{E}[\pi(\hat{q}, D)]$ is calculated with the estimated quantities $\hat{\mu}$, $\hat{\sigma}$, and $\hat{q}$.

As a direct consequence of these observations, we expect the EP_{ratio} to deviate from the theoretical optimum, and that this deviation will be greater for larger population standard deviations and smaller demand sample sizes (more uncertain estimates of $\hat{\mu}$, $\hat{\sigma}$, and $\hat{q}$).

6.2 Simulation results

The results are illustrated in Figure 4 and quantified in Table 4 for $N = 10$ (here too, for brevity, but more details are present in the code).

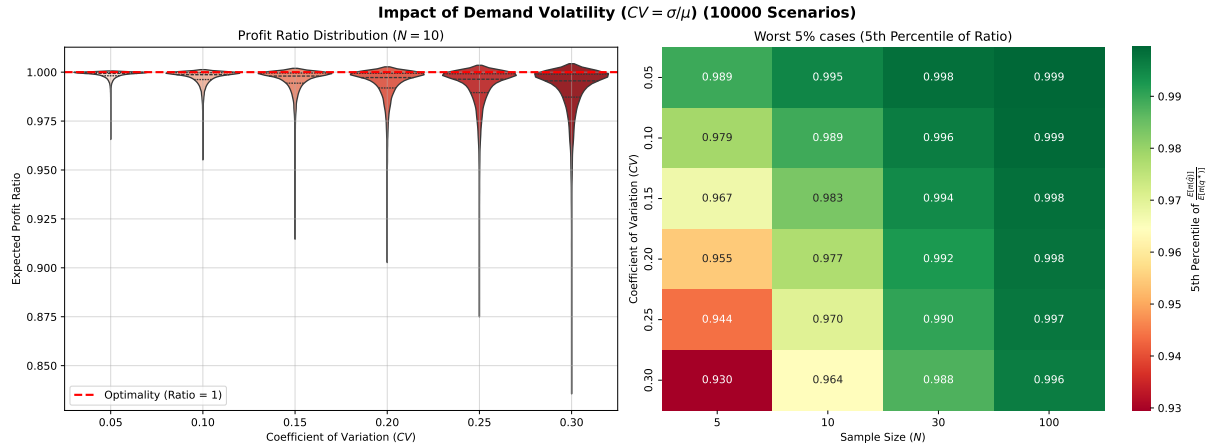


Figure 4: Distribution of the profit ratio as CV varies ($N = 10$) (left). Heat map of the 5th percentile as CV and N vary (right).

CV	Mean	Std	Median	2.5%	97.5%	95% CI Width
0.05	0.9986	0.0019	0.9994	0.9933	1.0000	0.0067
0.10	0.9972	0.0040	0.9987	0.9858	1.0000	0.0142
0.15	0.9957	0.0062	0.9980	0.9780	1.0000	0.0220
0.20	0.9939	0.0086	0.9972	0.9699	1.0000	0.0301
0.25	0.9921	0.0111	0.9964	0.9619	1.0000	0.0381
0.30	0.9904	0.0136	0.9956	0.9519	1.0000	0.0481

Table 4: Summary statistics of the impact of the coefficient of variation on the expected profit ratio ($N = 10$).

We comment on the results below:

- From the violin plot, the distributions show the usual negative asymmetry, with the mass concentrated near 1 and increasingly pronounced left tails as CV increases. For $N = 10$ and $CV = 0.30$, a relatively small sample in a condition of high demand volatility, in 2.5% of cases we obtain an expected profit less than 95% of the theoretical one.
- From the heat map, we find confirmation regarding the theoretical observation made, as we observe (both numerically and from the color gradient) that the largest changes in the EP_{ratio} are mainly

attributable to the sample size N , and consequently to parametric uncertainty. Consider the worst case of $N = 5$ and $CV = 0.30$: in 5% of cases we obtain profits at least 7% lower than the theoretical ones with an $EP_{ratio} = 0.930$

- Decreasing the CV and keeping N fixed, the EP_{ratio} grows up to 0.989
- Increasing N and keeping CV fixed, the EP_{ratio} grows up to 0.996

This testifies to the impact of the sample size.

7 Conclusions

Throughout this work, we have analyzed, using Monte Carlo simulations, how different sources of uncertainty influence the estimation of the newsvendor model's optimal quantity, and the consequent expected profit, using as a reference metric the ratio

$$EP_{ratio} = \frac{\mathbb{E}[\pi(\hat{q}, D)]}{\mathbb{E}[\pi(q^*, D)]}.$$

Under the Normal demand hypothesis, the dominant risk source turned out to be the parametric uncertainty on demand: small sample sizes produce unstable estimates of $(\hat{\mu}, \hat{\sigma})$, with pronounced left tails in the profit ratio distribution. The effect attenuates rapidly as the sample size increases: already for $N = 30$, the average degradation of the expected profit is less than 0.2% compared to the theoretical optimum.

The uncertainty on the salvage price $\hat{r}$ instead shows a significantly more contained impact. The main effect we observed concerns the appearance of right tails in the estimated quantities $\hat{q}$ distribution, due to the possibility of underestimating the overage cost.

The analysis of the $\rho = c_u/c_o$ ratio further highlights how the system's sensitivity to parametric uncertainty heavily depends on the economic context. Under the Normal demand hypothesis, low-margin products ($\rho < 1$) turn out to be structurally more fragile, with expected profit losses that in the worst cases exceed 30%.

Finally, a more volatile demand (high CV) amplifies the effect of estimation errors, although remaining secondary compared to the impact of the sample size.